

Nikhil Hegde

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EDUCATION

- **Nitte Meenakshi Institute of Technology** Bengaluru
Bachelor of Engineering in Computer Science; CGPA: 8.9 *July 2013 – May 2017*

PROGRAMMING SKILLS

- **Languages:** Python, Go, SQL **Tools and Frameworks:** Django, FastAPI, AWS, Kafka, Redis, Docker

EXPERIENCE

- **Inmar Technologies** Remote
Senior Software Engineer *September 2024 - Present*
 - **Offer Hub:** An AI powered management platform created for CPG and retailers to publish the offers digitally and manage them.
 - * Partnered with Data Science and AI teams to seamlessly integrate predictive models into production pipelines, optimizing backend workflows utilizing FastAPI and Django
 - * Championed Test-Driven Development (TDD) practices across the engineering team, authoring comprehensive unit and integration test suites that significantly elevated code quality and reduced regression defects
 - * Optimized high-throughput data processing and caching layers by leveraging AWS, Kafka, Redis, and Docker to ensure scalable and reliable feature deployment
 - * Optimized relational database performance by restructuring complex SQL queries, introducing efficient indexing, and managing connection pooling for PostgreSQL and MySQL workloads
 - * Leveraged AI-assisted development tools (e.g., GitHub Copilot/Cursor) to accelerate feature prototyping and automate the generation of robust TDD test suites, reducing time-to-market for new features by 25%

Tech Stack: Python, Django, FastAPI, AWS, PostgreSQL, MySQL, Redis, Kafka, Docker

- **Biofourmis India Pvt Ltd** Bengaluru
Senior Software Engineer *July 2021 - September 2024*
 - **Titration Engine:** An AI powered application specifically designed for chronic patients who are diagnosed with Hypertension, Heart failure, Lipids, Diabetes, Atrial fibrillation.
 - * Architected the migration of 'Titration Engine'—a critical, AI-powered chronic disease management application—from a legacy monolithic structure into high-performance microservices, driving a 50% boost in system performance
 - * Designed and implemented asynchronous message queues utilizing Kafka to handle bulk data processing, ensuring robust, low-latency data handling for thousands of patient profiles
 - * Instituted robust CI/CD and testing protocols using Jenkins, Docker, and comprehensive automated unit/integration tests to guarantee zero-downtime deployments
 - * Collaborated cross-functionally with AI and Data Science teams to integrate complex medical algorithms and machine learning models into clinical production environments
 - * Led and mentored a high-performing team of 3 engineers, establishing rigorous code review standards and industry best practices to streamline sprint deliveries
 - * Established distributed tracing and logging across the newly architected microservices, reducing the Mean Time to Detection (MTTD) for production bottlenecks
 - * Configured Redis as a highly efficient caching layer for frequently accessed clinical configuration data, reducing database read loads by 20%

Tech Stack: Python, Django, Flask, FastAPI, AWS, PostgreSQL, Redis, Jenkins, Kafka, Docker

- **Inflect Technologies** Bengaluru
Backend Engineer *April 2019 - July 2021*
 - **Video-R:** An AI powered application specifically designed for P&G (Procter & Gamble) to analyze video ads and provide insights related to user attention, brand attribution etc.
 - * Handled the application deployment inside P&G server and supported in adding 3 new features.
 - * Worked with AI team to integrate their models.

- **Curation:** An internal application created to curate the supermarket shelf images to generate insights related to brand, form, SKU (stock keeping unit) and export this information to AI team for training their models.
 - * Designed the entire backend and handled the deployment in AWS.
 - * Engineered real-time data pipelines to sync application data with Elasticsearch, enabling business stakeholders to generate complex analytical reports with sub-second latency
 - * Built custom Kibana dashboards to monitor infrastructure health, application error rates, and user behavior metrics, improving proactive system maintenance.

Tech Stack: Python, Django, AWS, PostgreSQL, MySQL, Redis, Jenkins, ElasticSearch, Docker

- **Ajir Technologies** Bengaluru
Backend Engineer *November 2017 - March 2019*

- **Zone-91:** A Management application created by Ajir Technologies for maintaining large gated societies.
 - * Designed the entire backend and handled the deployment in AWS.
 - * Wrote unit test cases and set up the CI/CD pipelines for automation.

Tech Stack: Python, Django, AWS, PostgreSQL, MySQL, Redis, Jenkins, Git, Docker

PUBLICATIONS

- Presented an IEEE paper regarding “Video Shot Detection in Transform Domain” in 2nd International Conference for Convergence in Technology (I2CT) - <https://ieeexplore.ieee.org/document/8226113>